Homework 4.3-4.4 #1

Student Name: ____________________________

Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **12**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 12

Use a calculator to evaluate each expression.
Round your answers to the nearest thousandth.
Do not round any intermediate computations.

$$\ln 23.4 =$$

$$\ln \frac{5}{6} =$$

Question 2 of 12

Rewrite each equation as requested.

(a) $\square^{\square} = \square$

(b) $\log_{\square} \square = \square$

(a) Rewrite as an exponential equation.

$$\log_7 \frac{1}{49} = -2$$

$$\square^{\square} = \square$$

(b) Rewrite as a logarithmic equation.

$$9^0 = 1$$

$$\log_{\square} \square = \square$$

Question 3 of 12

Rewrite each equation as requested.

(a) Rewrite as a logarithmic equation.

$$e^{y} = 6$$

Logarithmic equation: ______________

(b) Rewrite as an exponential equation.

$$\ln x = 9$$

Exponential equation: ______________

Question 4 of 12

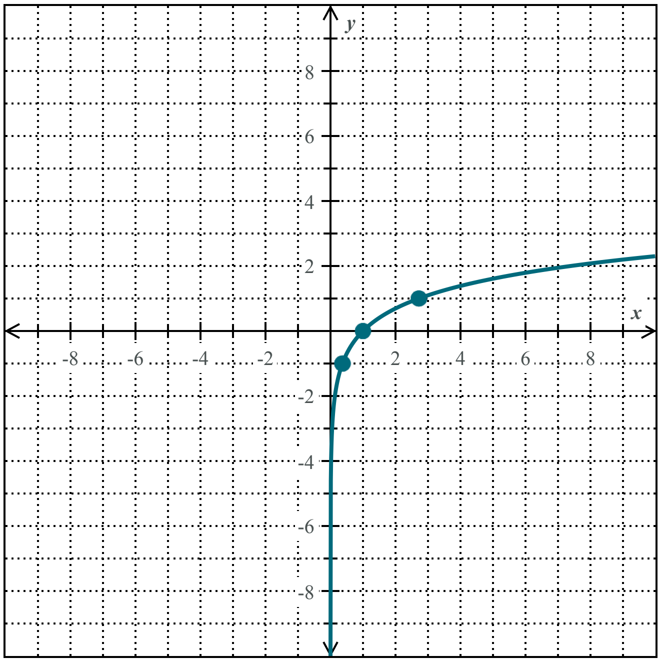
Evaluate each expression.

(a) $\log_8 8 = \square$

(b) $\log_3 \dfrac{1}{81} = \square$

Question 5 of 12

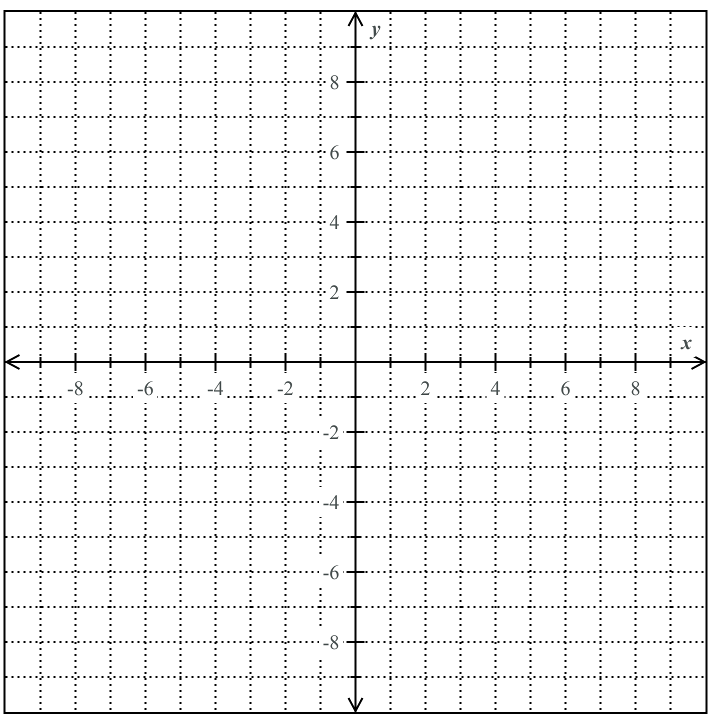
Below is the graph of $y = \ln x$. Translate it to become the graph of $y = \ln(x+1) + 3$.



Question 6 of 12

Graph the logarithmic function.

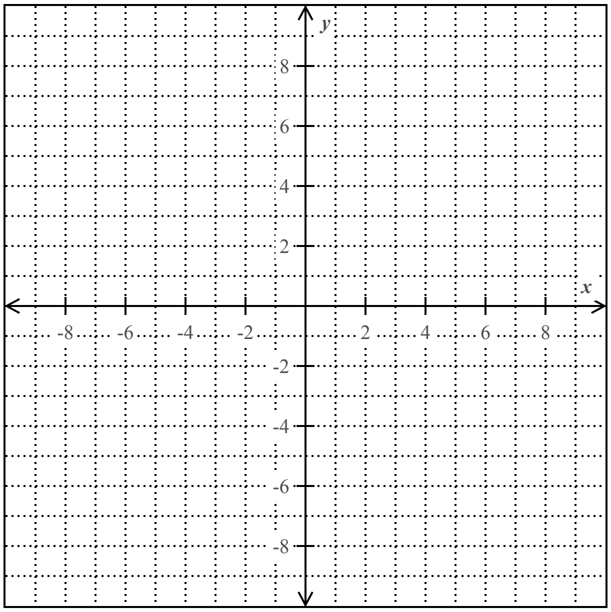
$$g(x) = -3 + \log_3 x$$



Question 7 of 12

(a) Graph the logarithmic function $g(x) = \log_3(x+3)$.

To do this, plot two points on the graph of the function, and also draw the asymptote. Then, click on the graph-a-function button.



(b) Give the domain and range of the function using interval notation.

Domain: ____________

Range: ____________

Question 8 of 12

Find the domain of the function.

$$f(x) = \log_2\left(\frac{-3}{x-6}\right)$$

Write your answer as an interval or union of intervals.

Question 9 of 12

Use the properties of logarithms to expand $\log\left(yz^{9}\right)$.

Each logarithm should involve only one variable and should not have any exponents or fractions. Assume that all variables are positive.

Question 10 of 12

Use the properties of logarithms to expand the following expression.

$$\log\sqrt[3]{\frac{xy^{4}}{z^{2}}}$$

Each logarithm should involve only one variable and should not have any radicals or exponents.

You may assume that all variables are positive.

$$\log\sqrt[3]{\frac{xy^{4}}{z^{2}}} = \square$$

Question 11 of 12

Write the expression as a single logarithm.

$$4\log_{m}(y-2) - 2\log_{m}(y+4)$$

$\log_{\_\_\_\_\_\_}\left(\_\_\_\_\_\_\right)$

Question 12 of 12

Use the change of base formula to compute $\log_{6}4$.

Round your answer to the nearest thousandth.

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Question 1 of 12

$\ln 23.4 = 3.153$

$\ln \frac{5}{6} = -0.182$

Question 2 of 12

(a) $7^{-2} = \frac{1}{49}$

(b) $\log_9 1 = 0$

Question 3 of 12

(a) Logarithmic equation: $\ln 6 = y$

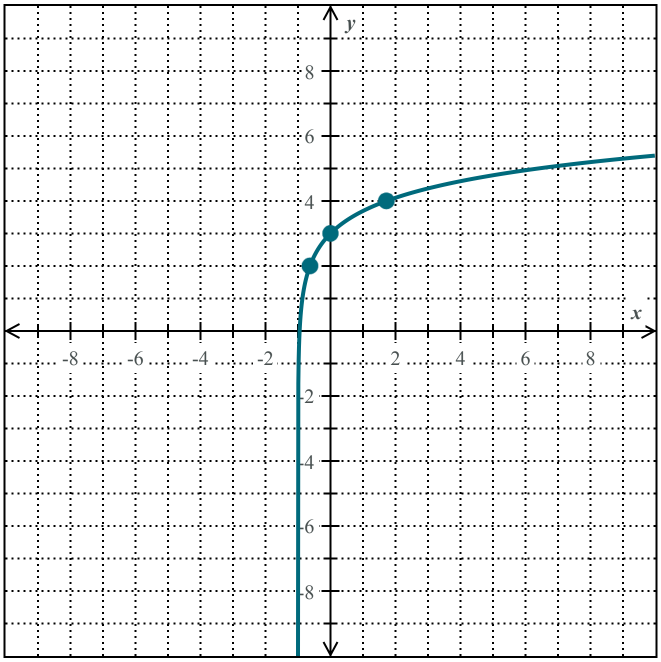
(b) Exponential equation: $e^9 = x$

Question 4 of 12

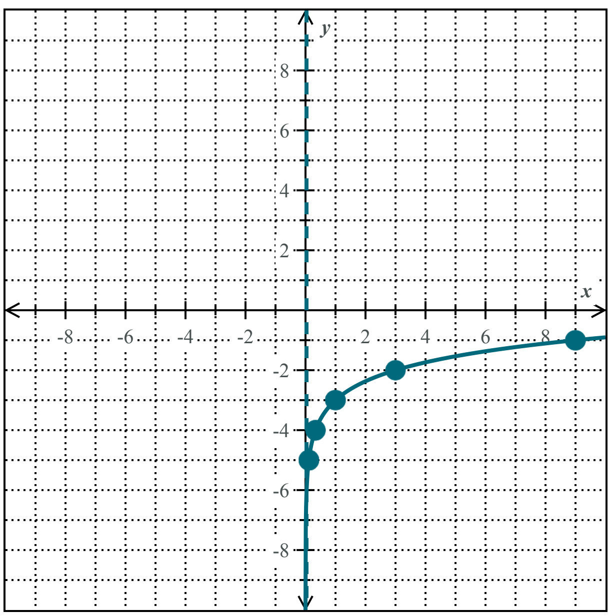
(a) $\log_8 8 = 1$

(b) $\log_3 \frac{1}{81} = -4$

Question 5 of 12

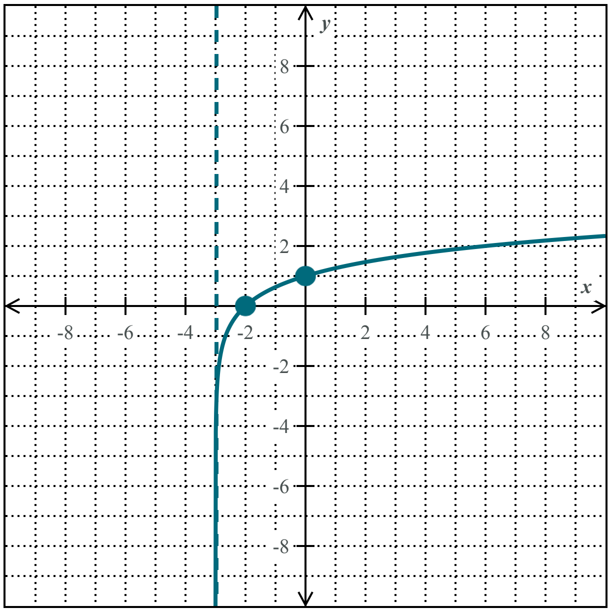


Question 6 of 12



Question 7 of 12

(a)



(b)

Domain: $(-3, \infty)$

Range: $(-\infty, \infty)$

Question 8 of 12

Domain: $(-\infty,\ 6)$

Question 9 of 12

$$\log\left(yz^9\right) = \log y + 9\log z$$

Question 10 of 12

$$\log\sqrt[3]{\frac{xy^4}{z^2}} = \frac{1}{3}\log x + \frac{4}{3}\log y - \frac{2}{3}\log z$$

Question 11 of 12

$$\log_m\left(\frac{(y-2)^4}{(y+4)^2}\right)$$

Question 12 of 12

0.774